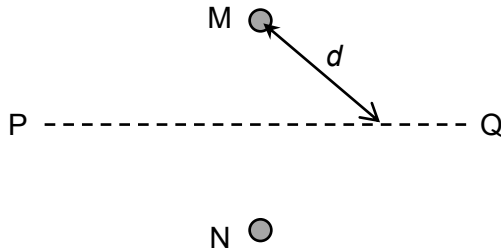


11 B



Gravitational potential ϕ is a **scalar** quantity. It is always negative and for point masses follow the formula $\phi = -\frac{GM}{r}$. So in this case, the resultant potential is $-2\frac{GM}{d}$. As such it is most negative (but not infinitely negative) at the midpoint, where it is nearest to both the stars, and less negative along either side as they move further away from the stars.
(modified from 2014 P1 Q13)

12 B

$$pV = nRT$$